SUPPORTED BY

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

ENERGY DEPARTMENT
Government of Rajasthan

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

AIDA
Collaboration | Innovation | Excellence

DISTRIBUTION UTILITY MEET
DUM 2026

ORGANIZER
ISGF
India Smart Grid Forum

KNOWLEDGE PARTNER
Power Foundation of India

HOST UTILITIES
JVVNL
Jodhpur Vidyut Vitran Nigam Limited
Ajmer Vidyut Vitran Nigam Limited

POWERED BY
Pelocal

RE PARTNER
RREC

CO - HOST UTILITIES
TATA POWER
Chandigarh Power Distribution Limited (CPDL)
TATA POWER-DDL
BSES Rajdhani Power Limited
NPCL
BSES Yamuna Power Limited

DUM 2026 CONFERENCE SESSION

Role of RE in Managing New and Emerging Loads and Ensuring Supply Reliability and Power Quality

Managing Uncertainties in California

Presented By

Siva Gunda, Vice Chair, California Energy Commission

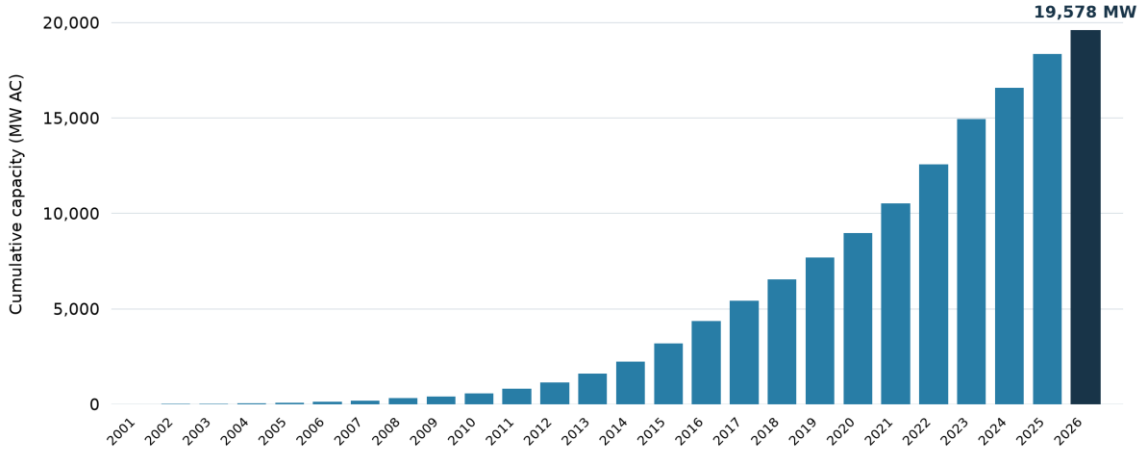
Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



Rapid Growth in Behind-the-Meter Solar and Battery Storage Capacity in California

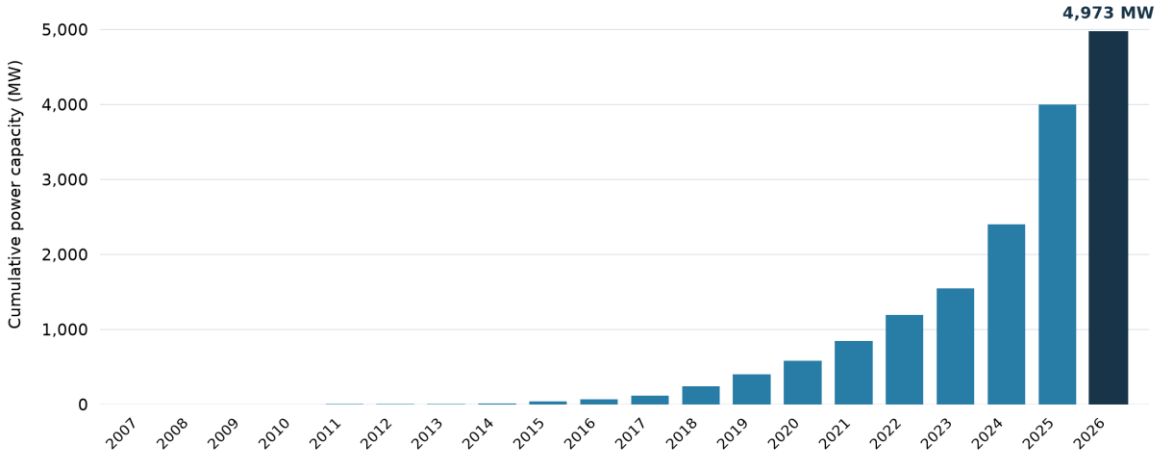
Cumulative BTM solar installed capacity

PG&E, SCE and SDG&E service territories | Through August 2026



Cumulative BTM battery storage capacity

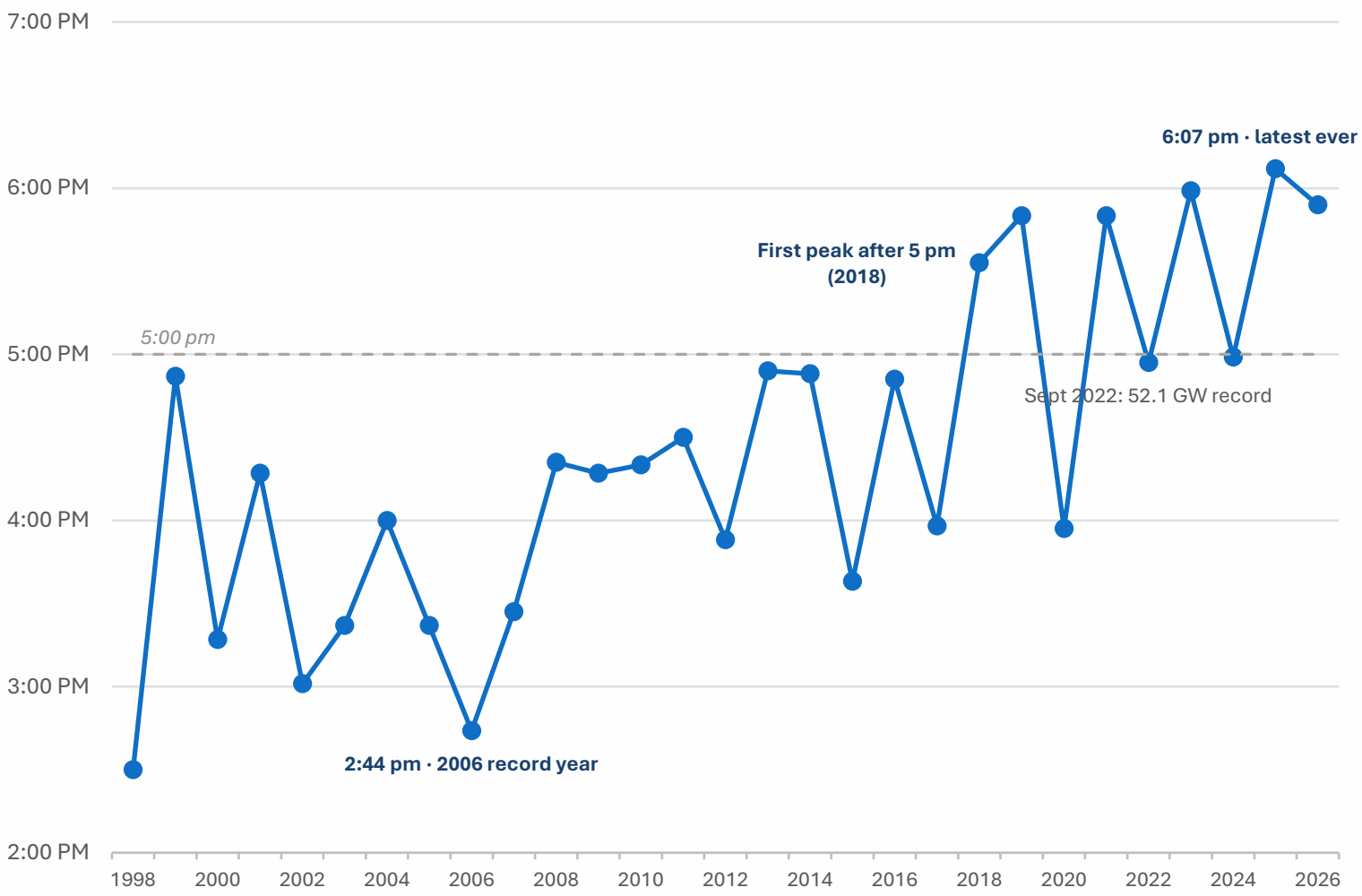
Residential and commercial combined | 2026 partial year



Accumulated by COD year. 2026: residential through June; commercial through July.

The Peak Has Moved Into the Evening Ramp

Time of day of CAISO's annual system peak, 1998–2026 (net of rooftop solar)



What fills the evening ramp

Rooftop solar moved the system peak. CAISO's recorded demand is already net of behind-the-meter solar, which hollows out midday load. Average peak time: 3:35 pm (1998–2005) → 5:27 pm (2019–2026).

The net-load peak is later still. After utility-scale solar and wind are netted out, the peak the fleet must actually serve falls around 7–8 pm — after sunset.

Gas capacity stayed; its job changed. Fleet capacity is roughly flat while its capacity factor fell from 44% to 22%. Gas now runs for ramping and evening hours rather than bulk energy.

Storage takes the first hours. Batteries charged on midday solar now carry the early ramp; gas, imports and hydro carry the later hours and multi-day heat events like September 2022.

Planning implication. Reliability is measured at the net-load peak, not the system peak — which is why California values duration, flexibility and demand-side resources, and keeps a reserve.

Source: CAISO Peak Load History (1998–2025, instantaneous peak) and CAISO Key Statistics, Sept 2026 (2026 peak, 1-minute average; as of Oct 13, 2026). System demand is net of behind-the-meter generation and gross of utility-scale solar and wind. CAISO balancing area ≈ 80% of California load.

Rooftop Solar Compensation: Net Energy Metering to Net Billing Tarrif

NEM 1.0 · 1996–2016

Full retail credit for every exported kWh

- Export at noon credited at ~retail (today 30–40¢) when grid value was 2–5¢.
- Solar customers paid almost nothing toward wires, wildfire, programs — while using the grid every evening.
- 20-year grandfathering locked the terms in.

NEM 2.0 · 2016–2023

Retail credit kept; small fixes added

- Mandatory time-of-use rates; one-time interconnection fee.
- Non-bypassable charges (~2–3¢/kWh) on imports for public programs.
- Core problem untouched: retail rates kept rising, export value kept falling. ~3× more installs than NEM 1.0.

Net Billing Tariff · 2023–

Exports paid at grid value, by hour

- Export credit = avoided cost: near zero midday, high on late-summer evenings; ~75% below retail on average.
- Imports billed at retail TOU. Economics now favour solar + storage and self-consumption over export.
- Installs fell sharply in year one, then recovered with far higher battery attachment. 9-year legacy for NEM 2.0.

$8.5 billion / yr

estimated cost shift to non-solar customers by end-2024, up from $3.4 B in 2021

≈ 16–23% of a non-solar household's bill (CPUC Public Advocates Office)

How it is measured. Cost shift = bill savings granted to NEM customers − avoided costs (energy, capacity, GHG, grid). NEM 2.0 ≈ $5.7 B, NEM 1.0 ≈ $2.6 B, Net Billing ≈ $0.25 B of the 2024 total.

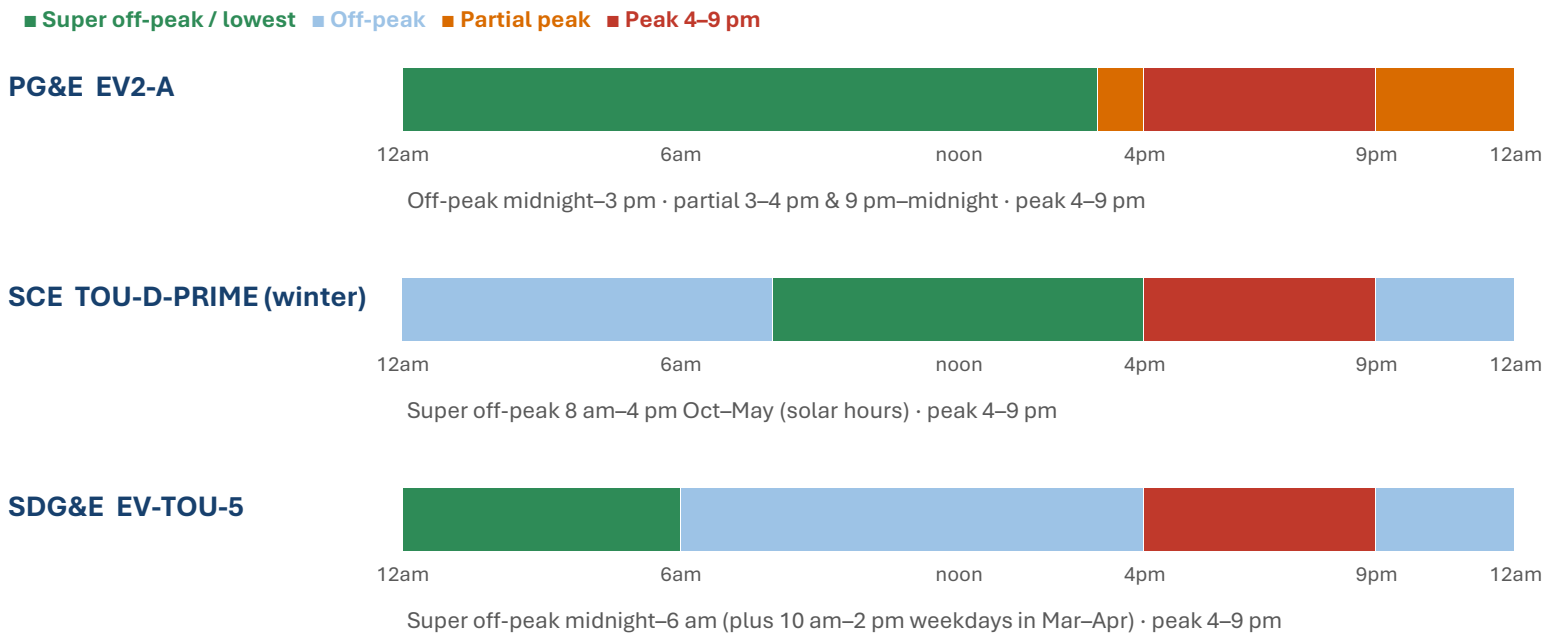
Why it is sticky. Legacy terms protect ~1.5 million NEM 1/2 systems for 20 years from interconnection; the shift persists into the 2030s even though new customers are on grid-value pricing. The income-graduated fixed charge (2025–26) addresses fixed-cost recovery more broadly.

Lesson. *Set export compensation at grid value and by hour from the start, and settle fixed-cost recovery before penetration is high — reforming a promise made to a million customers is far harder than designing it well.*

Sources: CPUC Decision 22-12-056 (Net Billing Tariff); CPUC Public Advocates Office, NEM cost-shift fact sheet (Aug 2024) and 2024 analysis; CPUC NEM program data. Cost-shift estimate is contested by solar industry; methodology is the CPUC's avoided-cost framework.

EV Tariffs Work: Price Signals Move Charging Off the 4–9 pm Peak

HOW THE THREE IOU EV RATES SHAPE THE DAY (residential, weekday)



Common design: all three put the highest price on 4–9 pm when solar fades and the net-load peak arrives; SCE and SDG&E add a midday window to absorb solar. **Limit:** fixed windows mean every timer starts at the same minute — see next slide.

MEASURED RESPONSE

~12%

lower household demand during 4–9 pm on CAISO's five highest-load days

≈ 12.4 MW across ~60,000 SDG&E households on EV time-of-use rates (2024), vs. a matched control group

Scale it up. PG&E's infrastructure study: orchestrated EV flexibility can cut the 5 pm coincident peak by 2.2–3.1 GW by 2040.

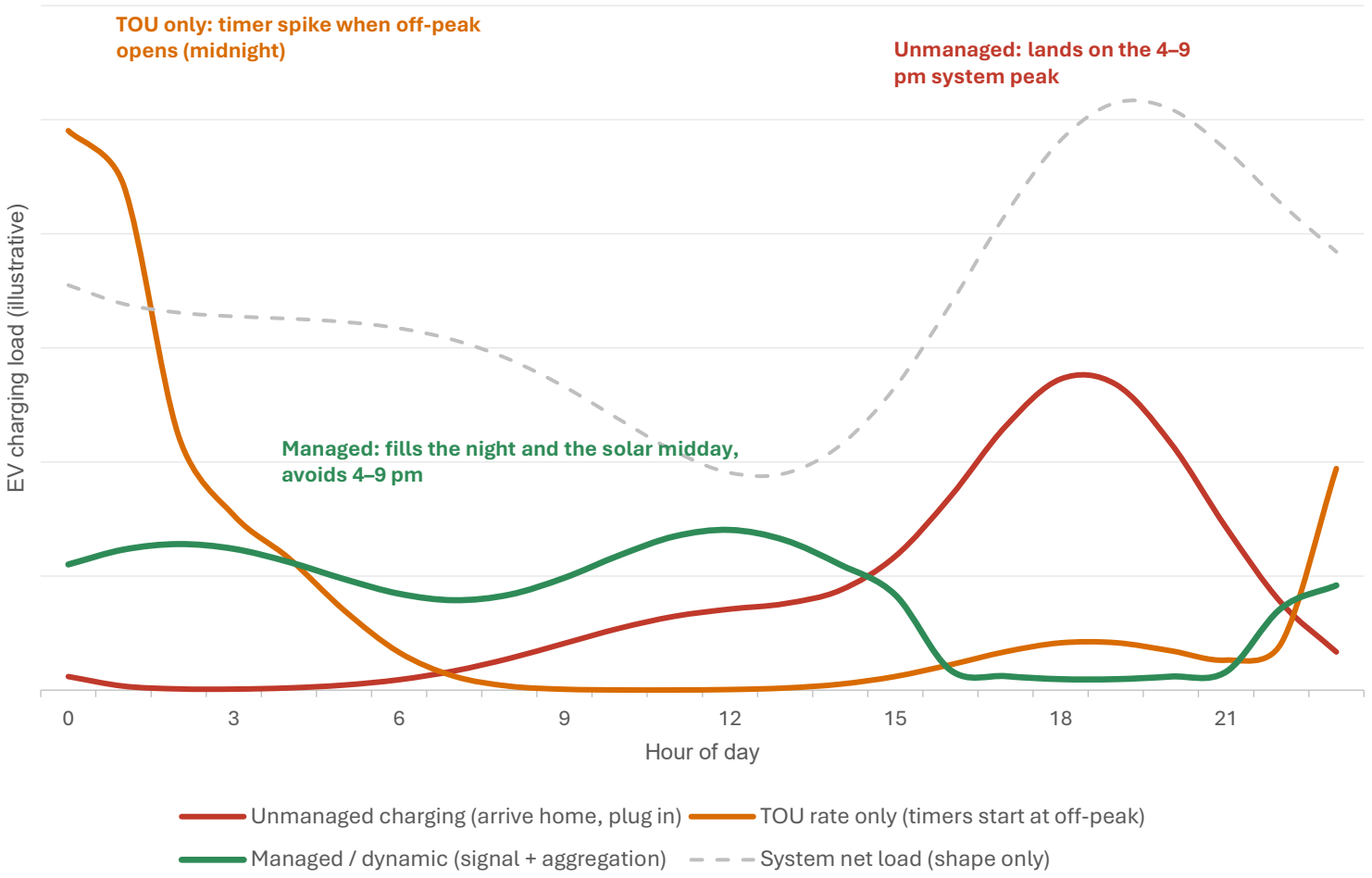
Default matters. Residential customers are on TOU by default; EV-specific rates deepen the price spread (peak vs. lowest ≈ 2–3×).

Caveat. The 12.4 MW is total household load reduction, not metered EV charging alone; it is the response to price, before any managed-charging signal.

Next frontier. Dynamic prices and automated managed charging layered on TOU, so EVs respond to real-time grid and local-feeder conditions.

Sources: PG&E EV2-A, SCE TOU-D-PRIME, SDG&E EV-TOU-5 tariff sheets (windows simplified; seasons and weekends vary); SDG&E PY2024 EV time-varying-rate load-impact evaluation (Demand Side Analytics); PG&E Electric Infrastructure Study (2026).

Time-of-Use Moves the EV Peak; Managed Charging Removes It



Curves are illustrative shapes, not measured data. Sources: CPUC Vehicle-Grid Integration Forum (Mar 2026); CEC 2025 IEPR transportation hourly results (Oct 2025); PG&E Electric Infrastructure Study.

What California has learned

Price alone creates a new peak. Under TOU EV rates, chargers start together when the off-peak window opens; PG&E's distribution study found TOU raised peaks on more than 20% of feeders (E3 / Integral Analytics, 2026).

A signal fixes it. CEC EPIC pilot (~1,800 EVs): load balancing consistently removed timer peaks vs. a TOU counterfactual. SCE ChargeSmart and PG&E hourly flex pricing (−$0.04 to $1.09/kWh) do the same at scale.

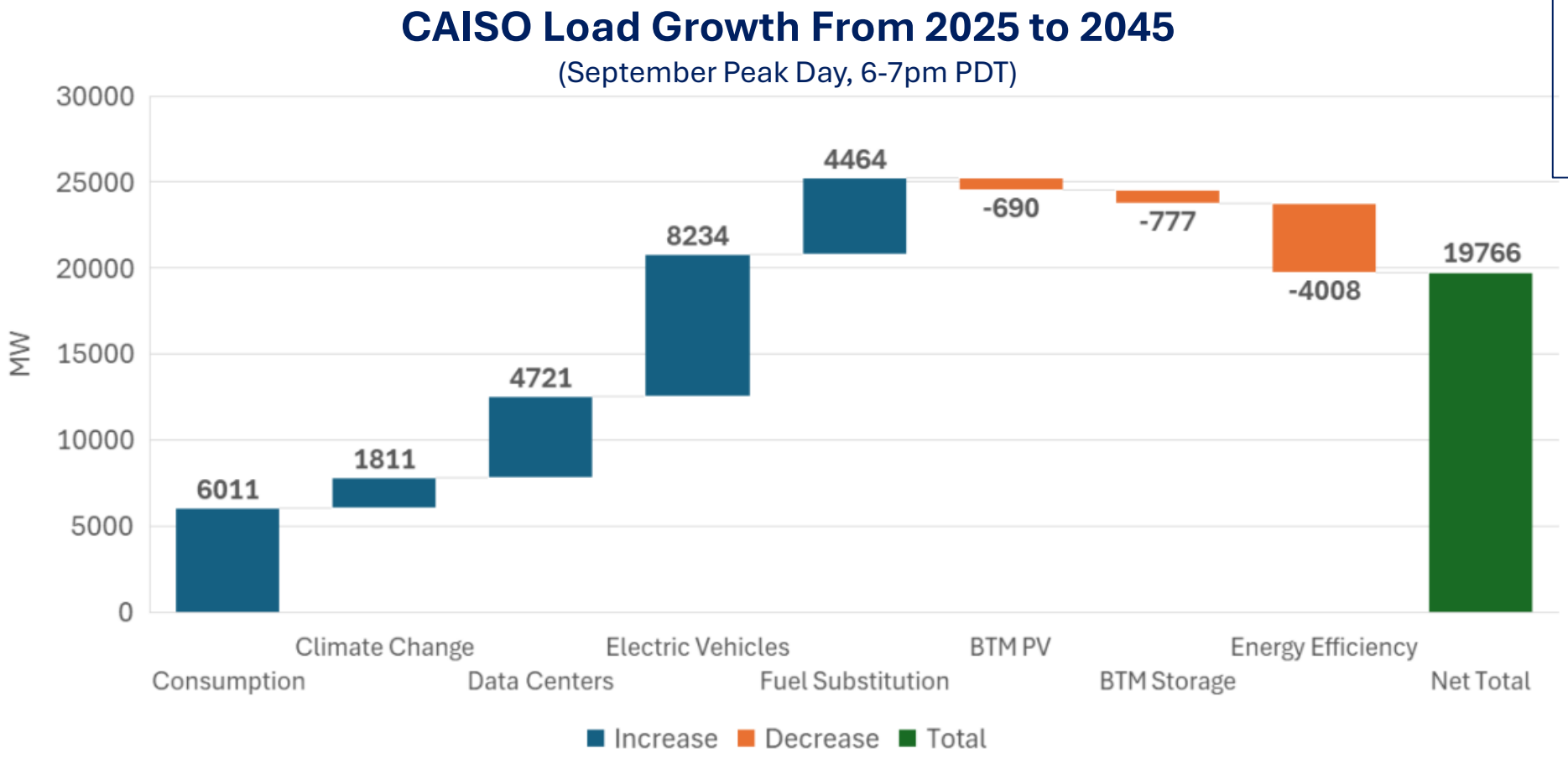
The prize is at the system peak. PG&E estimates orchestrated flexibility cuts the 5 pm coincident peak by 2.2–3.1 GW and distribution costs by ~$1.8 B through 2040 — only if orchestrated, not merely time-priced.

Forecast it. The IEPR EV load model now takes TOU rates and participation as inputs; the 2026 update adds observed AMI charging shapes.

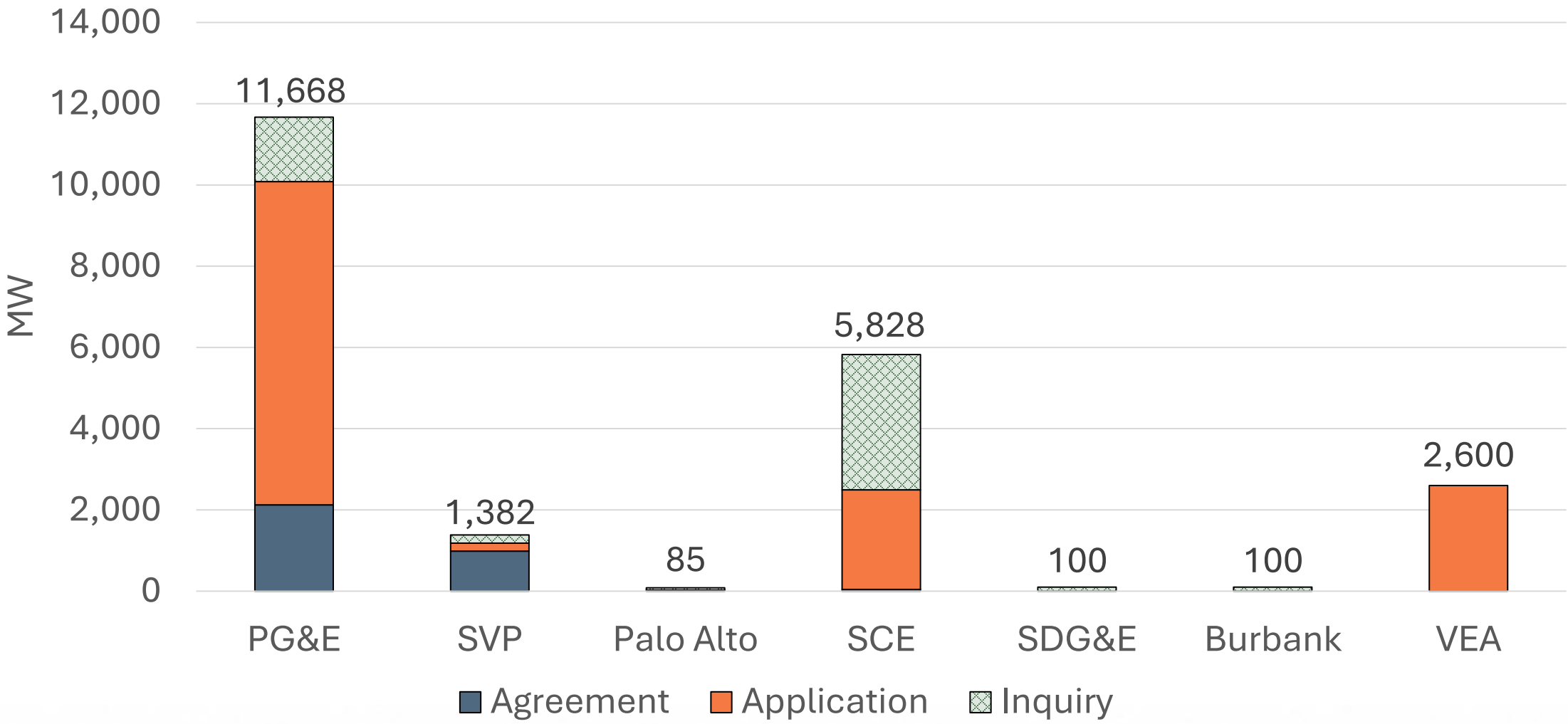
State Forecasting Significant Load Growth

EVs and data centers both meaningful drivers of growth

- Net Load 2025 46.5 GW
- Net Load 2045 66.0 GW
- Increase 19.5 GW
- EV Portion 42 %
- Data Center Portion 24 %



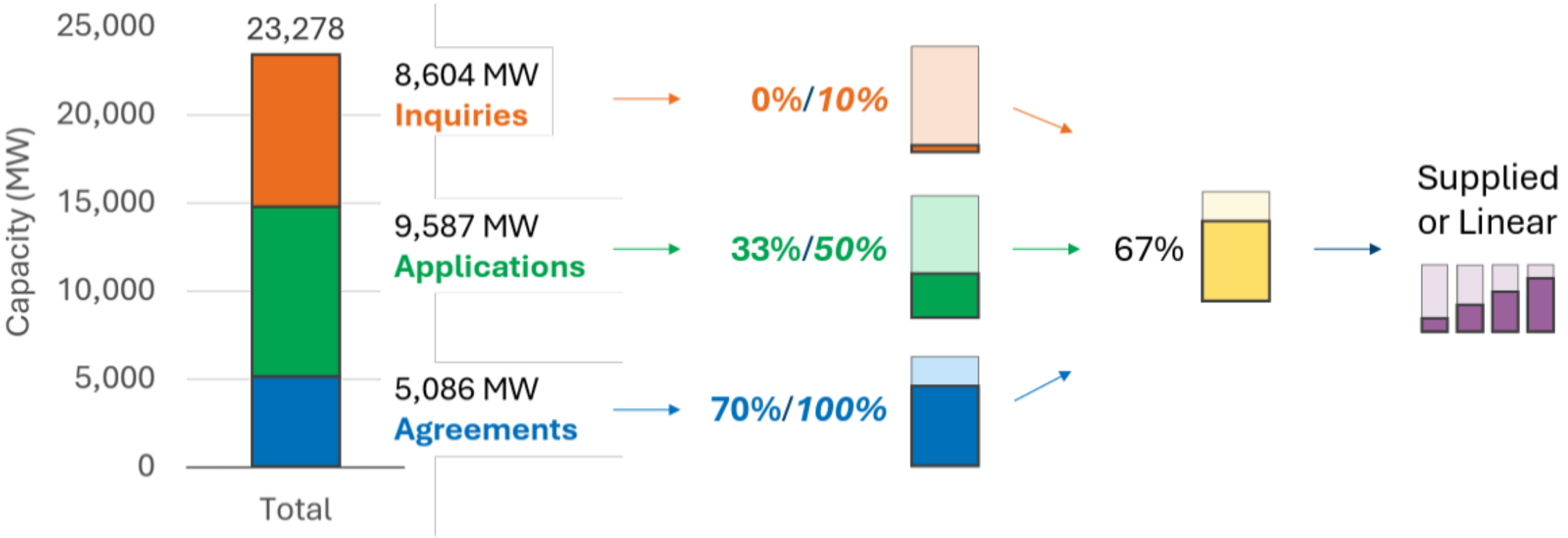
2025 Data: Capacity Requested



Source: CEC with data from each utility

Assumptions Applied to the 2025 IEPR Forecast

Data Center Capacity Requests (Dec 2025, MW) x **Confidence Levels** (mid/*high*) x **Utilization Factor** x **Ramp Rate**



Source: CEC Staff

Data Centers: Rate Reducer or Cost Shifter? The Tariff Decides

THE OPPORTUNITY

Flat, high-load-factor demand can spread fixed costs

Volumetric rates fall when new sales exceed new costs. Every kWh sold above marginal cost pays down the wires, wildfire and program costs everyone shares.

Scale is real. PG&E's data-center pipeline ≈ 10 GW; PG&E estimates ≥1% lower bills for all customers per 1 GW served under the right rules.

Headroom exists if loads flex. US grid could absorb ~76 GW of new load with only 0.25% annual curtailment (Duke, 2025) — the ramp hours are the constraint, not the energy.

THE RISK

Speculative, inflexible or short-lived loads shift costs to everyone else

Stranded investment. Generation and transmission built for a load that never arrives, ramps slowly, or leaves — paid by remaining customers for decades.

Below-cost contracts. Confidential special contracts can price large loads under cost of service; other classes absorb the gap (Harvard EELP, 2025).

Peak coincidence. A 24/7 load that cannot curtail adds capacity need at the evening net-load peak — the most expensive hour to serve.

EMERGING BEST PRACTICE

55 US large-load tariffs surveyed (LBNL, Aug 2026) converge on:

Pay for what you reserve. Minimum bills at ~80% of contracted demand; 10–12-year terms; exit fees; collateral (e.g., $1.5M/MW).

Separate class, hold-harmless test. Define by size and load factor, not industry; regulator must show existing customers are not worse off (Xcel CO, NV Energy).

Reward flexibility and clean self-supply. Curtailment credits, staged ramps, behind-the-meter zero-carbon resources, customer-chosen clean portfolios (NV Clean Transition Tariff).

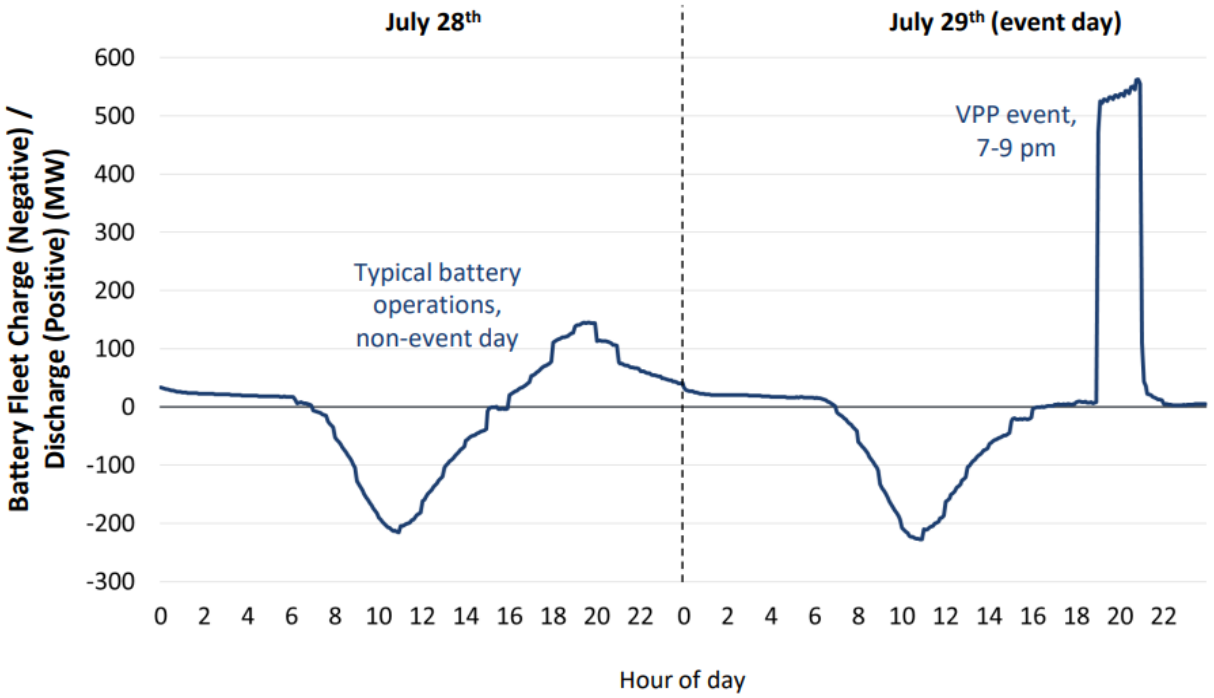
CALIFORNIA'S ANSWER (SB 886 / AB 2383, signed Sept 2026) CPUC must adopt data-center interconnection and generation tariffs by Jan 2028: loads ≥ threshold (≤25 MW) pay incremental generation and transmission costs for ≥10 years, with collateral, non-completion and early-exit fees sized to hold other ratepayers harmless, and credit for behind-the-meter zero-carbon resources. Rate design in R.26-04-009; PG&E Rule 30 (transmission-level interconnection) pending.

Sources: LBNL, Electricity Rate Designs for Large Loads (Aug 2026); RMI Electricity Affordability Toolkit; Duke Nicholas Institute, Rethinking Load Growth (2025); Harvard EELP, Extracting Profits from the Public (2025); E3 (2025); PG&E Assembly testimony (Jan 2026); Stoel Rives summary of SB 886/AB 2383.

Nation’s Largest Virtual Power Plant

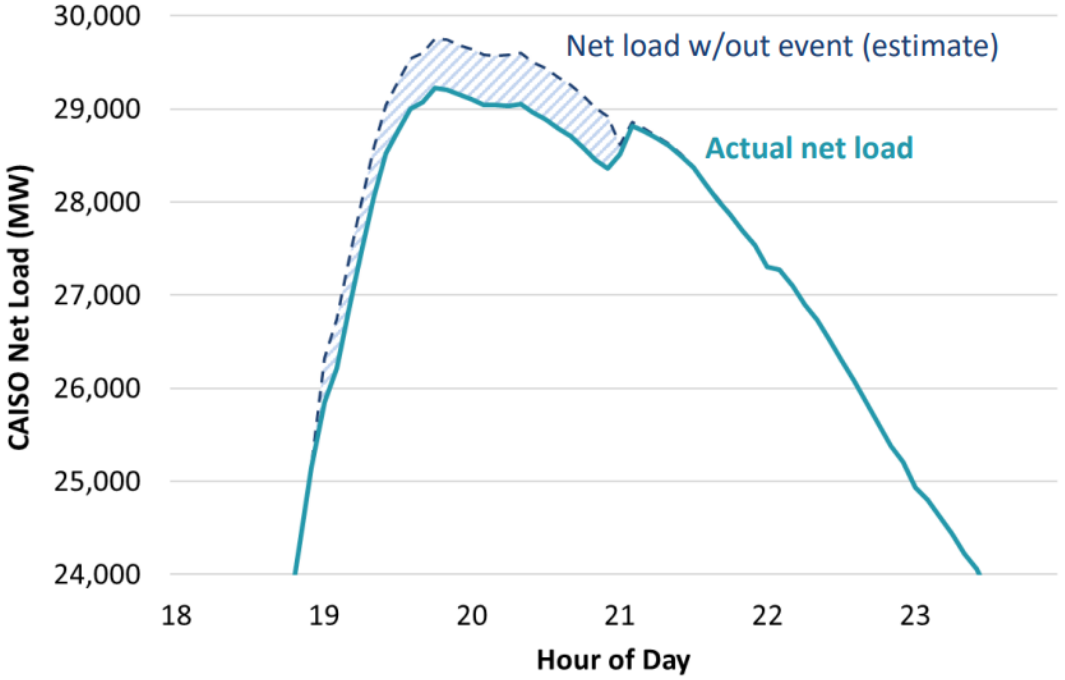
Over 1 GW enrolled in Demand Side Grid Support (DSGS) program

Battery Operations Before and During the Event



Notes: Based on Brattle analysis of 5-minute telemetry data provided by Sunrun and Tesla. Battery dispatch is raw power output, without any baseline adjustments.

CAISO System Net Load on Event Day



Notes: Net load sourced from CAISO and reflects actual demand less solar and wind output. Baseline net load in the absence of the event was constructed using 5-minute telemetry data provided by Sunrun and Tesla. All battery output is shown as a reduction in net load.

Source: Brattle

THANK YOU

For discussions/suggestions/queries email: ***dum@indiasmartgrid.org***

www.dumindia.in

Links/References (If any)

